Embedded System Lab Manual

Subject Name: Embedded System Lab

Subject Code: 7IT06

Experiment No.: 5

Title: Demonstrate OLED Display with Arduino Uno for Text and Sensor Data Visualization.

Objectives:

1. To interface an OLED display with Arduino Uno using I²C communication.
2. To display simple text messages on the OLED screen.
3. To acquire sensor data (e.g., temperature or analog input) and visualize it on the OLED display.

Required Components/Kit:

- Arduino Uno Board
- USB Cable, Breadboard and Jumper Wires
- Computer with Arduino IDE Installed
- OLED Display Module (0.96-inch, SSD1306, I²C interface)
- DHT11 Temperature & Humidity Sensor (or Potentiometer as analog sensor)

Libraries Required:

- Adafruit_GFX.h
- Adafruit_SSD1306.h

Theory:

OLED (Organic Light Emitting Diode) is a self-lighting display technology with high contrast and low power consumption. Most Arduino-compatible OLEDs use the SSD1306 driver with I²C communication (SDA, SCL).

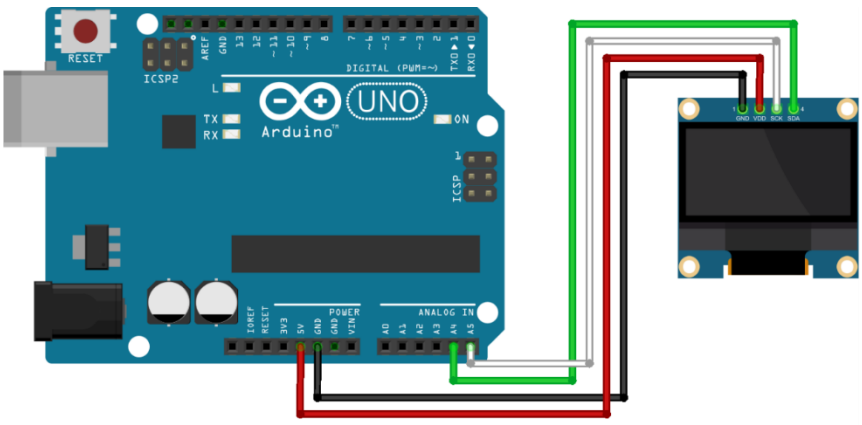
The Arduino Uno (ATmega328P) has built-in support for I²C on A4 (SDA) and A5 (SCL) pins. A sensor (e.g., DHT11) can provide real-world data (temperature, humidity, or analog values) which can be read by Arduino and displayed on the OLED.

Procedure:

1. Connect Arduino Uno to PC via USB cable.
2. Wire the OLED display to Arduino using I²C (A4, A5).
3. Connect the DHT11 sensor (or potentiometer).
4. Open Arduino IDE and install required libraries:
    1. Adafruit SSD1306
    2. Adafruit GFX
    3. DHT Sensor Library
5. Copy the above program into the IDE.
6. Select Board: Arduino Uno and the correct Port.
7. Compile and Upload the code.
8. Observe the OLED displaying both text and real-time sensor readings.

Circuit Diagram (Description):

1. Connect OLED SDA → Arduino A4, OLED SCL → Arduino A5.
2. Connect OLED VCC → 5V and GND → GND.
3. Connect DHT11 sensor data pin → Arduino D2 with 10kΩ pull-up resistor (or Potentiometer wiper → A0).



Code:

```
// Experiment 5: OLED Display with Arduino Uno
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
```

```
#include <DHT.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET    -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

// DHT11 Sensor Setup
#define DHTPIN 2                                        // Data pin connection
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

void setup()
{
Serial.begin(9600);
if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))          // Initialize OLED
        {
                Serial.println(F("OLED initialization failed"));
                for(;;);
        }
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);

// Display welcome message
display.setCursor(0, 0);
display.println("Embedded Systems Lab");
display.println("Experiment 5: OLED");
display.display();
delay(2000);
dht.begin();
}
```

```
void loop()
{
        float temp = dht.readTemperature();             // Read sensor data
        float hum = dht.readHumidity();
        display.clearDisplay();                         // Clear display
        display.setCursor(0, 0);
        display.println("OLED + Sensor Demo");          // Show text and sensor values
        display.print("Temp: ");
        display.print(temp);
        display.println(" C");
        display.print("Hum : ");
        display.print(hum);
        display.println(" %");
        display.display();
        delay(2000);
}
```

Observation Table:

Sr. No	Sensor Input (Temp °C / Humidity %)	OLED Display Output
1	28 °C, 60 %	Temp: 28 °C, Hum: 60 %
2	30 °C, 65 %	Temp: 30 °C, Hum: 65 %

Result:

The OLED display was successfully interfaced with Arduino Uno. Static text and live sensor data were displayed, demonstrating real-time visualization capability.

Conclusion:

This experiment demonstrated the interfacing and controlling of an output device (LED) using a push button.